

Primary 4-6

TEACHER PRACTICAL ACTIVITY MANUAL

Implementation guide for PA-201 to PA-300

OBSERVE -> MEASURE -> INVESTIGATE -> ANALYZE -> EXPLAIN

Purpose	Turn the GrowMeal practical cards into safe, repeatable, age-appropriate teaching sessions.
Teacher role	Prepare the site, demonstrate technique, supervise safety, question learners, check evidence and connect the activity to the journal, mission cards and assessment system.
Core cycle	PREPARE -> DEMONSTRATE -> PRACTISE -> RECORD -> DISCUSS -> CONNECT -> CLEAN UP
Assessment	Use observation, student records and the linked practical/investigation evidence. Avoid grading recall alone.
Local adaptation	Substitute locally available materials when the learning objective, safety and measurement quality are preserved.

WHOLE-MANUAL SAFETY & IMPLEMENTATION STANDARD

Teachers must inspect the garden before each activity; establish boundaries; match tools to learner age; supervise water, cutting tools, electrical equipment, pumps, nutrient solutions and instruments; require handwashing after garden work; prohibit tasting unless a separate approved food-safety activity explicitly permits it; stop any activity when conditions are unsafe.

Before class	Check site, weather/working conditions, tools, labels, water, access routes and learner grouping.
During class	Demonstrate first; keep learners within assigned zones; monitor tool use and protect plants, structures and instruments.
After class	Count/clean/store tools; close water systems; restore labels; record damage, abnormal readings or incidents; wash hands.
Research integrity	Never invent, overwrite or silently delete observations. Flag questionable data and document repeat/reference checks.
Accessibility	Use peer roles, large-print/visual instructions, reachable observation stations and alternative recording methods where appropriate.

GARDEN 01 - Raised Food Garden

Teaching focus: raised-bed growth, spacing and care.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Raised Food Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-201 - OBSERVE - Raised Food Garden

Learning objective	Guide pupils to observe a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-202 - MEASURE - Raised Food Garden

Learning objective	Guide pupils to measure a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-203 - QUESTION - Raised Food Garden

Learning objective	Guide pupils to question a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-204 - SAMPLE - Raised Food Garden

Learning objective	Guide pupils to sample a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-205 - INVESTIGATE - Raised Food Garden

Learning objective	Guide pupils to investigate a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-206 - CALCULATE - Raised Food Garden

Learning objective	Guide pupils to calculate a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-207 - GRAPH - Raised Food Garden

Learning objective	Guide pupils to graph a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-208 - ANALYZE - Raised Food Garden

Learning objective	Guide pupils to analyze a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-209 - EXPLAIN - Raised Food Garden

Learning objective	Guide pupils to explain a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-210 - RECOMMEND - Raised Food Garden

Learning objective	Guide pupils to recommend a question about raised-bed growth, spacing and care, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Raised Food Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 02 - Geometry Garden

Teaching focus: shapes, sides, corners, length and garden layout.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Geometry Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-211 - OBSERVE - Geometry Garden

Learning objective	Guide pupils to observe a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-212 - MEASURE - Geometry Garden

Learning objective	Guide pupils to measure a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-213 - QUESTION - Geometry Garden

Learning objective	Guide pupils to question a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-214 - SAMPLE - Geometry Garden

Learning objective	Guide pupils to sample a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-215 - INVESTIGATE - Geometry Garden

Learning objective	Guide pupils to investigate a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-216 - CALCULATE - Geometry Garden

Learning objective	Guide pupils to calculate a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-217 - GRAPH - Geometry Garden

Learning objective	Guide pupils to graph a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-218 - ANALYZE - Geometry Garden

Learning objective	Guide pupils to analyze a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-219 - EXPLAIN - Geometry Garden

Learning objective	Guide pupils to explain a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-220 - RECOMMEND - Geometry Garden

Learning objective	Guide pupils to recommend a question about shapes, sides, corners, length and garden layout, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Geometry Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 03 - Grid Experiment Garden

Teaching focus: fair comparisons, treatments, controls and replication.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Grid Experiment Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-221 - OBSERVE - Grid Experiment Garden

Learning objective	Guide pupils to observe a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-222 - MEASURE - Grid Experiment Garden

Learning objective	Guide pupils to measure a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-223 - QUESTION - Grid Experiment Garden

Learning objective	Guide pupils to question a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-224 - SAMPLE - Grid Experiment Garden

Learning objective	Guide pupils to sample a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-225 - INVESTIGATE - Grid Experiment Garden

Learning objective	Guide pupils to investigate a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-226 - CALCULATE - Grid Experiment Garden

Learning objective	Guide pupils to calculate a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-227 - GRAPH - Grid Experiment Garden

Learning objective	Guide pupils to graph a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-228 - ANALYZE - Grid Experiment Garden

Learning objective	Guide pupils to analyze a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-229 - EXPLAIN - Grid Experiment Garden

Learning objective	Guide pupils to explain a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-230 - RECOMMEND - Grid Experiment Garden

Learning objective	Guide pupils to recommend a question about fair comparisons, treatments, controls and replication, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Grid Experiment Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 04 - Keyhole Garden

Teaching focus: access, bed geometry, moisture and composting.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Keyhole Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-231 - OBSERVE - Keyhole Garden

Learning objective	Guide pupils to observe a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-232 - MEASURE - Keyhole Garden

Learning objective	Guide pupils to measure a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-233 - QUESTION - Keyhole Garden

Learning objective	Guide pupils to question a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-234 - SAMPLE - Keyhole Garden

Learning objective	Guide pupils to sample a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-235 - INVESTIGATE - Keyhole Garden

Learning objective	Guide pupils to investigate a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-236 - CALCULATE - Keyhole Garden

Learning objective	Guide pupils to calculate a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-237 - GRAPH - Keyhole Garden

Learning objective	Guide pupils to graph a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-238 - ANALYZE - Keyhole Garden

Learning objective	Guide pupils to analyze a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-239 - EXPLAIN - Keyhole Garden

Learning objective	Guide pupils to explain a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-240 - RECOMMEND - Keyhole Garden

Learning objective	Guide pupils to recommend a question about access, bed geometry, moisture and composting, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Keyhole Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 05 - Compost Garden

Teaching focus: temperature, moisture, decomposition and maturity.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Compost Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-241 - OBSERVE - Compost Garden

Learning objective	Guide pupils to observe a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-242 - MEASURE - Compost Garden

Learning objective	Guide pupils to measure a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-243 - QUESTION - Compost Garden

Learning objective	Guide pupils to question a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-244 - SAMPLE - Compost Garden

Learning objective	Guide pupils to sample a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-245 - INVESTIGATE - Compost Garden

Learning objective	Guide pupils to investigate a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-246 - CALCULATE - Compost Garden

Learning objective	Guide pupils to calculate a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-247 - GRAPH - Compost Garden

Learning objective	Guide pupils to graph a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-248 - ANALYZE - Compost Garden

Learning objective	Guide pupils to analyze a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-249 - EXPLAIN - Compost Garden

Learning objective	Guide pupils to explain a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-250 - RECOMMEND - Compost Garden

Learning objective	Guide pupils to recommend a question about temperature, moisture, decomposition and maturity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Compost Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 06 - Gravity Drip Garden

Teaching focus: flow, uniformity and water delivery.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Gravity Drip Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-251 - OBSERVE - Gravity Drip Garden

Learning objective	Guide pupils to observe a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-252 - MEASURE - Gravity Drip Garden

Learning objective	Guide pupils to measure a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-253 - QUESTION - Gravity Drip Garden

Learning objective	Guide pupils to question a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-254 - SAMPLE - Gravity Drip Garden

Learning objective	Guide pupils to sample a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-255 - INVESTIGATE - Gravity Drip Garden

Learning objective	Guide pupils to investigate a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-256 - CALCULATE - Gravity Drip Garden

Learning objective	Guide pupils to calculate a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-257 - GRAPH - Gravity Drip Garden

Learning objective	Guide pupils to graph a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-258 - ANALYZE - Gravity Drip Garden

Learning objective	Guide pupils to analyze a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-259 - EXPLAIN - Gravity Drip Garden

Learning objective	Guide pupils to explain a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-260 - RECOMMEND - Gravity Drip Garden

Learning objective	Guide pupils to recommend a question about flow, uniformity and water delivery, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Gravity Drip Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 07 - Vertical Garden

Teaching focus: multi-level growth, light and water distribution.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Vertical Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-261 - OBSERVE - Vertical Garden

Learning objective	Guide pupils to observe a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-262 - MEASURE - Vertical Garden

Learning objective	Guide pupils to measure a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-263 - QUESTION - Vertical Garden

Learning objective	Guide pupils to question a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-264 - SAMPLE - Vertical Garden

Learning objective	Guide pupils to sample a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-265 - INVESTIGATE - Vertical Garden

Learning objective	Guide pupils to investigate a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-266 - CALCULATE - Vertical Garden

Learning objective	Guide pupils to calculate a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-267 - GRAPH - Vertical Garden

Learning objective	Guide pupils to graph a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-268 - ANALYZE - Vertical Garden

Learning objective	Guide pupils to analyze a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-269 - EXPLAIN - Vertical Garden

Learning objective	Guide pupils to explain a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-270 - RECOMMEND - Vertical Garden

Learning objective	Guide pupils to recommend a question about multi-level growth, light and water distribution, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Vertical Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 08 - Crop Rotation Garden

Teaching focus: crop families, sequence and soil cover.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Crop Rotation Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-271 - OBSERVE - Crop Rotation Garden

Learning objective	Guide pupils to observe a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-272 - MEASURE - Crop Rotation Garden

Learning objective	Guide pupils to measure a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-273 - QUESTION - Crop Rotation Garden

Learning objective	Guide pupils to question a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-274 - SAMPLE - Crop Rotation Garden

Learning objective	Guide pupils to sample a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-275 - INVESTIGATE - Crop Rotation Garden

Learning objective	Guide pupils to investigate a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-276 - CALCULATE - Crop Rotation Garden

Learning objective	Guide pupils to calculate a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-277 - GRAPH - Crop Rotation Garden

Learning objective	Guide pupils to graph a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-278 - ANALYZE - Crop Rotation Garden

Learning objective	Guide pupils to analyze a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-279 - EXPLAIN - Crop Rotation Garden

Learning objective	Guide pupils to explain a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-280 - RECOMMEND - Crop Rotation Garden

Learning objective	Guide pupils to recommend a question about crop families, sequence and soil cover, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Crop Rotation Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 09 - Water-Smart Garden

Teaching focus: infiltration, irrigation, runoff and water productivity.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Water-Smart Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-281 - OBSERVE - Water-Smart Garden

Learning objective	Guide pupils to observe a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-282 - MEASURE - Water-Smart Garden

Learning objective	Guide pupils to measure a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-283 - QUESTION - Water-Smart Garden

Learning objective	Guide pupils to question a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-284 - SAMPLE - Water-Smart Garden

Learning objective	Guide pupils to sample a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-285 - INVESTIGATE - Water-Smart Garden

Learning objective	Guide pupils to investigate a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-286 - CALCULATE - Water-Smart Garden

Learning objective	Guide pupils to calculate a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-287 - GRAPH - Water-Smart Garden

Learning objective	Guide pupils to graph a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-288 - ANALYZE - Water-Smart Garden

Learning objective	Guide pupils to analyze a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-289 - EXPLAIN - Water-Smart Garden

Learning objective	Guide pupils to explain a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-290 - RECOMMEND - Water-Smart Garden

Learning objective	Guide pupils to recommend a question about infiltration, irrigation, runoff and water productivity, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Water-Smart Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 10 - Intro Soilless Garden

Teaching focus: growing media, flow, pH and EC.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Intro Soilless Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-291 - OBSERVE - Intro Soilless Garden

Learning objective	Guide pupils to observe a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils observe using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-292 - MEASURE - Intro Soilless Garden

Learning objective	Guide pupils to measure a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils measure using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-293 - QUESTION - Intro Soilless Garden

Learning objective	Guide pupils to question a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils question using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-294 - SAMPLE - Intro Soilless Garden

Learning objective	Guide pupils to sample a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils sample using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-295 - INVESTIGATE - Intro Soilless Garden

Learning objective	Guide pupils to investigate a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils investigate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-296 - CALCULATE - Intro Soilless Garden

Learning objective	Guide pupils to calculate a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils calculate using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-297 - GRAPH - Intro Soilless Garden

Learning objective	Guide pupils to graph a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils graph using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-298 - ANALYZE - Intro Soilless Garden

Learning objective	Guide pupils to analyze a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils analyze using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-299 - EXPLAIN - Intro Soilless Garden

Learning objective	Guide pupils to explain a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils explain using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-300 - RECOMMEND - Intro Soilless Garden

Learning objective	Guide pupils to recommend a question about growing media, flow, pH and EC, collect comparable evidence, analyze it and explain a defensible result.
Materials	Journal, ruler/tape, timer, measuring container, labels, calculator where useful, graph paper and teacher-approved garden instruments.
DIY / low-cost option	Use marked reused containers for volume, string grids for sampling, locally made stakes for labels and hand-drawn graph sheets.
Teacher preparation	Define the comparison, samples and safety boundary in the Intro Soilless Garden. Prepare labels and check measuring tools before class.
Step-by-step delivery	1. State the investigation question. 2. Identify what changes and what should stay the same. 3. Demonstrate measurement technique. 4. Pupils recommend using repeated samples. 5. Tabulate evidence. 6. Calculate or graph a useful summary. 7. Discuss pattern, anomaly and limitation. 8. Make one recommendation.
Questions to ask	Is the comparison fair? What pattern is visible? Is there an unusual value? What limitation matters? What does the evidence justify?
Expected evidence	Labeled table, units, calculation/graph, conclusion, limitation and recommendation.
Differentiation	Provide calculation scaffolds and partially completed tables; extension pupils calculate an additional rate, percentage or comparison.
Troubleshooting	If the test becomes unfair, stop data collection, identify the uncontrolled difference, correct it where possible and document the limitation.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

LEVEL IMPLEMENTATION REVIEW

Use this page after completing the level's practical sequence.

Strongest learner competency	
Competency needing reteaching	
Garden/system needing maintenance	
Materials to replenish	
Safety or accessibility improvement	
Next mission/assessment priority	